UNIVERSITY OF ALBERTA

Certifies by this document to all whom it may concern that

MENNATULLAH SIAM

having completed all the statutory requirements
has been granted the degree of

DOCTOR OF PHILOSOPHY

in Computing Science

and awarded all the rights and privileges pertaining to this degree.

Given at this university on the nineteenth day of November, two thousand and twenty-one
in the one hundred and fifteenth year of the University of Alberta.

CHANCELLOR

PRESIDENT

CHAIR OF THE BOARD



REGISTRAR